2.2 Inductive and Deductive Reasoning

Name _____ Period ____ Date _____

Geometry 2.2 · Student sheet

I can…

- use inductive reasoning to describe a pattern and make a conjecture.
- show a conjecture is false with a counterexample.
- use the Laws of Detachment and Syllogism to reach a conclusion.

Vocabulary · fill in the missing word

Conjecture — an unproven statement that is based on _____

Inductive reasoning — finding a pattern in specific cases and then writing a _____ for the general case

Counterexample — a specific case for which a _____ is false

Deductive reasoning — using facts, _____, accepted properties, and the laws of logic to form a logical argument

Part 1 · Discovery: is it always true?

Test each claim with four examples of your own.

Claim	My examples	Always true?
The sum of two odd numbers is even.		
The sum of three consecutive integers is 3 times the middle one.		
$n^2 + n + 41$ is always prime.		

One example that breaks a claim is a _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Inductive reasoning: look at examples, find a _____, and make a conjecture.

Example 1

Describe the pattern and find the next number: 3, 7, 11, 15, …

Example 2

Find the sum of several sets of three consecutive integers. Make a conjecture.

Counterexample: to show a conjecture is false, you need only _____ counterexample.

Example 3

Find a counterexample: “The difference of two positive numbers is always positive.”

Law of Detachment: if $p \rightarrow q$ is true and p is true, then _____ is true.

Example 4

What can you conclude? “If two segments have the same length, then they are congruent. AB = CD = 5.”

Law of Syllogism: if $p \rightarrow q$ and $q \rightarrow r$ are true, then _____ is true.

Example 5

Write the statement that follows: “If $x^2 > 25$, then $x^2 > 20$.” “If $x^2 > 20$, then $x^2 > 10$.”

Part 3 · Partner practice

Partner 1

Find a counterexample: “All prime numbers are odd.”

Partner 2

Write the statement that follows: “If a figure is a square, then it is a rectangle.” “If a figure is a rectangle, then it has four right angles.”